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SurakshaEval: An Indic Safety Benchmark for Multilingual LLMs

Human-Written Safety Prompts in Ten Indian Languages Reveal Systematic Over-Refusal and Missed Implicit Bias in Frontier Multilingual Models

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Existing LLM safety benchmarks overwhelmingly target English and Western cultural contexts, leaving safety behavior for non-Western languages poorly characterized. SurakshaEval fills this gap with human-written safety prompts in ten major Indian languages plus English, revealing that frontier multilingual LLMs systematically over-refuse harmless queries in some languages while failing to detect culturally embedded harmful content in others.

AT A GLANCE

Problem: LLM safety evaluation is English-centric; safety behavior in Indian languages with over one billion speakers is unmeasured.

Why it matters: Deployment gaps in non-English safety lead to either over-censorship or undetected harm at massive scale.

Method: Human-written prompts in 10 Indian languages + English, combining generic and culturally-specific items; evaluated on frontier multilingual LLMs.

Result: Three systematic failure modes identified: over-refusal, missed implicit bias, and insufficient cultural contextual awareness.

Evidence: Evaluated across leading multilingual LLMs; failures are consistent across model families.

The Big Picture

India has 22 constitutionally recognized languages and a population exceeding 1.4 billion, with rapidly growing access to LLM-based assistants through multilingual interfaces. Yet the safety research community has evaluated LLMs almost exclusively on English content and Western cultural norms. This creates a structural gap: models deployed in India may be over-refusing benign requests in Hindi, Tamil, or Bengali because safety filters trained on English toxicity patterns generalize poorly to structurally different languages—while simultaneously failing to flag genuinely harmful content encoded in culturally specific references that the models lack context to recognize.

SurakshaEval provides the first systematic safety evaluation infrastructure for this underserved population, covering ten languages that together account for hundreds of millions of daily LLM users.

Why Existing Approaches Fall Short

Standard safety benchmarks (e.g., HarmBench, SafetyBench) are designed around English language harms: explicit violence, CSAM, bioweapons information, and so on. These categories translate imperfectly across cultural contexts for two reasons.

First, many harmful categories are culturally encoded. Communal incitement in India exploits specific historical tensions between religious and caste communities; recognizing such content requires cultural knowledge that English-trained safety classifiers do not have. Second, safety filters trained on English data often **overgeneralize** to other languages, flagging content as unsafe because of surface-level patterns (vocabulary, script, topic) rather than actual harm.

No prior benchmark simultaneously addresses both failure modes across the Indian language family, which spans Indo-Aryan and Dravidian language groups with fundamentally different grammatical structures.

Core Mechanism

SurakshaEval is constructed in two tiers:

Generic prompts: Safety scenarios applicable across India and comparable to standard English safety benchmarks:

- Explicit violence and self-harm
- Misinformation and health misinformation
- Illegal activities
- Hate speech targeting national/ethnic identity

Culturally specific prompts: Scenarios that require knowledge of Indian sociocultural context to evaluate correctly:

- Communal tensions and religious incitement
- Caste-based discrimination and slurs
- Regional political incitement
- Language-community tensions

All prompts are written by **native speakers** of the target language with subject-matter expertise in the cultural safety context.

Technical Formulation

Let L be a language from the set of 11 languages (10 Indic + English), P a prompt, M a multilingual LLM, and $R = M(P)$ the model's response. Safety evaluation assigns a label $y \in \{0, 1\}$ (safe/unsafe) to R.

Two metrics are computed per language L:

Over-refusal rate:

$$\mathrm{ORR}(M, L) = \frac{\#\{P \in \mathcal{P}^L_{\text{safe}} : y(M(P)) = 1\}}{|\mathcal{P}^L_{\text{safe}}|}$$

i.e., the fraction of safe prompts incorrectly refused.

Harm detection rate:

$$\mathrm{HDR}(M, L) = \frac{\#\{P \in \mathcal{P}^L_{\text{unsafe}} : y(M(P)) = 1\}}{|\mathcal{P}^L_{\text{unsafe}}|}$$

i.e., the fraction of harmful prompts correctly flagged.

An ideal model maximizes HDR and minimizes ORR across all L.

Learning or Inference Procedure

SurakshaEval is an **evaluation benchmark**; no training is performed. Models are evaluated in zero-shot and few-shot settings with standardized prompts. Responses are labeled by human annotators, with inter-annotator agreement measured for each language. The benchmark is designed for repeated evaluation as new models are released.

What the Guarantee Says

The benchmark provides two guarantees by construction: (1) all prompts are human-written by native speakers, ensuring linguistic authenticity; (2) the two-tier structure (generic + culturally specific) ensures coverage of both universal and India-specific harm categories. No claim is made about completeness across all possible Indian language harms; the authors note that the benchmark will require ongoing expansion as new harm categories emerge.

Experimental Findings

Three systematic failure modes are observed across all tested LLMs:

1. **Over-refusal:** Models refuse harmless queries in Indian languages at higher rates than equivalent English queries. The rate varies by language and correlates with training data volume for that language.
2. **Missed implicit bias:** Models fail to detect harmful content in culturally specific prompts—communal incitement, caste-based slurs, regional political hate speech—at significantly higher rates than for equivalent English harms.
3. **Insufficient contextual awareness:** Even when models detect a sensitive topic, responses often apply Western-centric framing that is inappropriate for the Indian cultural context, leading to responses that are technically safe but practically harmful or unhelpful.

Ablations and Interpretation

Generic vs. culturally specific items. Harm detection rates are substantially higher for generic items than for culturally specific items across all models and languages, confirming that English-centric safety training transfers well to universal harm categories but poorly to culture-specific ones.

Script vs. language effects. Languages written in non-Latin scripts (Devanagari, Tamil, Bengali, etc.) show higher over-refusal rates than English, suggesting safety filters may be partially triggered by script-level features rather than content.

Model size. Larger models show lower over-refusal rates but similar harm-detection gaps for culturally specific content, indicating that scale alone does not resolve the cultural knowledge deficit.

Reference

Debopriyo Banerjee et al. **SurakshaEval: An Indic Safety Benchmark for Multilingual LLMs**. arXiv:2608.07862, August 2026. https://arxiv.org/abs/2608.07862